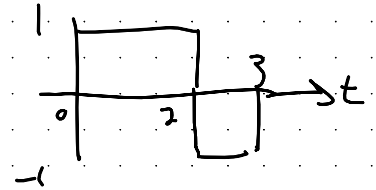
Assignment 1

SEE THE TEXTBOOK FOR THE FULL QUESTIONS.

1.1-1(a)

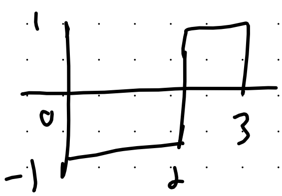
ENERGY OF



$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$$

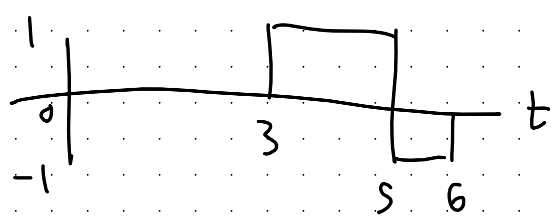
$$E = \int_0^2 (1)^2 \, dt + \int_2^3 |(-1)|^2 \, dt = 2 + 1 = 3$$

b)



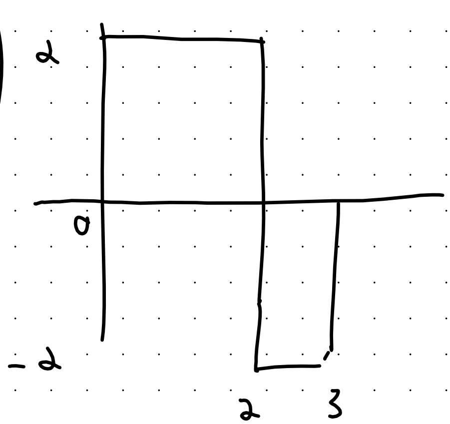
$$E = \int_0^2 |-1|^2 \, dt + \int_2^3 |1| \, dt = 2 + 1 = 3$$

c)



$$E = \int_3^5 |1|^2 \, dt + \int_5^6 |1|^2 \, dt = 2 + 1 = 3$$

d)



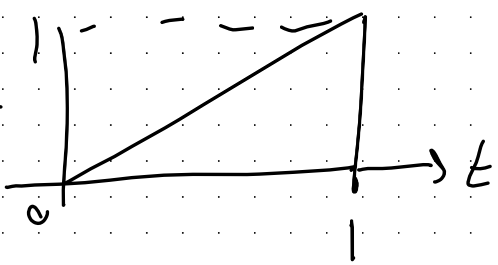
$$E = \int_0^2 (2)^2 \, dt + \int_2^3 |-2|^2 \, dt = 8 + 4 = 12$$

→ INVERTING DOESN'T CHANGE TOTAL ENERGY

→ MULTIPLYING BY K INCREASES ENERGY BY K^2

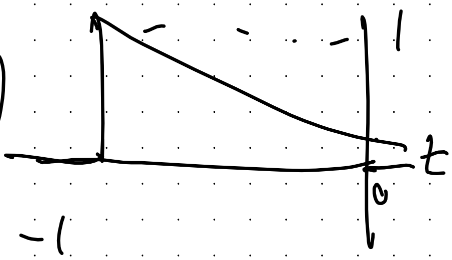
1.1-2

a)



$$E_x = \int_0^1 t^2 dt = \frac{1}{3} t^3 \Big|_0^1 = \frac{1}{3}$$

b)



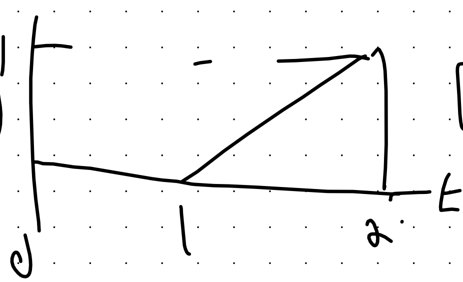
$$E_{x_1} = \int_{-1}^{0} |-t|^2 dt = \frac{1}{3} t^3 \Big|_{-1}^{0} = \frac{1}{3}$$

c)

$$E_{x_2} = \int_0^1 |-t|^2 dt = \frac{1}{3} t^3 \Big|_0^1 = \frac{1}{3}$$

LET X = t-1

d)



$$E_{x_3} = \int_1^2 (t-1)^2 = \int_0^1 x^2 dx = \frac{1}{3}$$

SHIFTS DON'T CHANGE ENERGY

e)

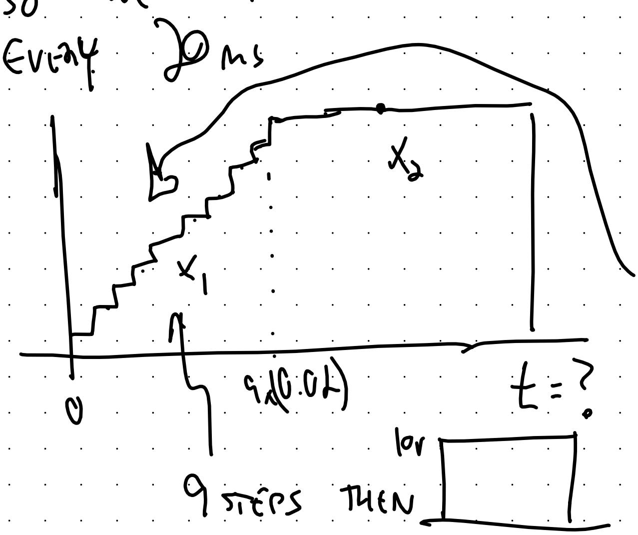
$$E_{x_4}(t) = \int_0^1 |2t|^2 dt = \frac{4}{3} t^3 \Big|_0^1 = \frac{4}{3}$$

1.1-5

ORIGINAL SYSTEM OUTPUTS 10V PULSE FOR 3 SECONDS

BUT WE WANT A "SOFT START"

SO WE WANT SOMETHING THAT STEPS UP TO 10 VOLTS IN 1V INCREMENTS EVERY 20 ms



FIND t SO THAT BOTH HAVE THE SAME ENERGY

HAVE TO MAKE UP FOR THE START!

FAULHABER'S FORMULA FROM A TABLE

$$E_{x_1} = \sum_{i=1}^{9} i^2 (0.02) = 0.02\left[\frac{9(9+1)(2(9)+1)}{6}\right] = 5.7$$

$$E_{x_2} = (t - 0.18)(10^2) = 100T - 18 \quad \Big\} \; E_{TOTAL} = 100T - 18 + 5.7$$

INTEGRATION AS AREA UNDER CURVE

$E_{ORIG} = E_{TOTAL}$

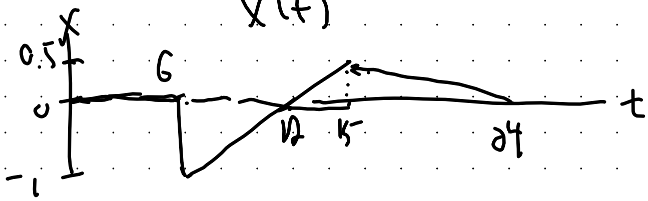
$3 \times 10^2 = 100T - 18 + 5.7$

$$T = \frac{300 - 5.7 + 18}{100} = 3.123 \text{ SECONDS}$$

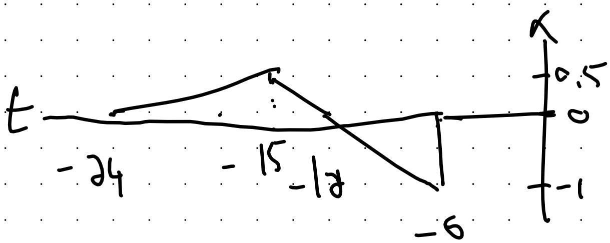
$\therefore$ THE SIGNAL DURATION OF THE SOFT START IS 3.123 s

1.2-1

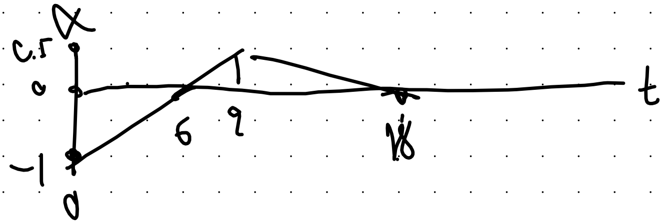
x(t)



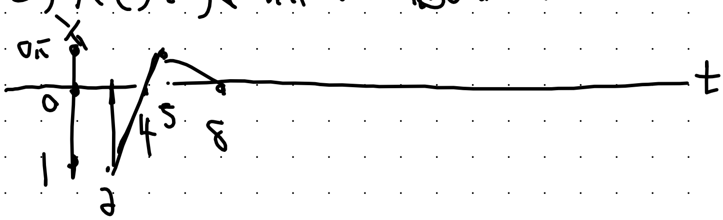
a) $x(-t)$ ← FLIP ABOUT X-AXIS



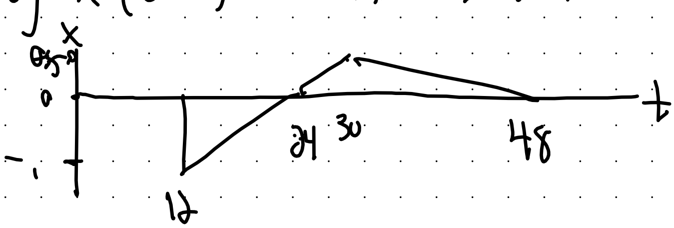
b) $x(t+6)$ ← SHIFT LEFT BY 6



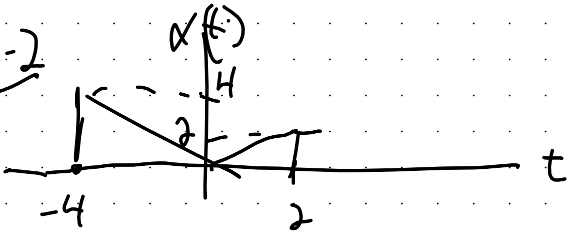
c) $X(3t)$ ← SHRINK ABOUT X-AXIS BY $1/3$

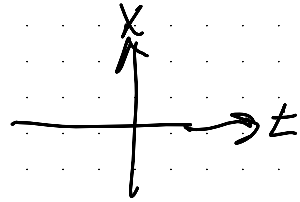


d) $x(t/2)$ ← EXPAND BY A FACTOR OF 2 ABOUT X-AXIS.



1.2-2



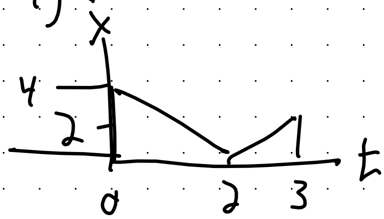


a) $x(t-4)$ ← SHIFT RIGHT BY 4

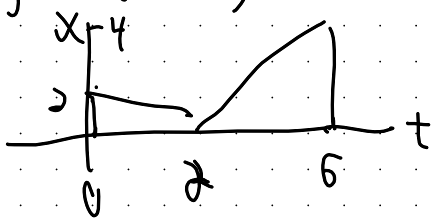
b) $x(t/1.5)$ ← EXPAND ABOUT X-AXIS BY A FACTOR OF 1.5

c) $X(-t)$ ← FLIP ABOUT X-AXIS

d) $x(2t-4)$ ← SHIFT RIGHT BY 4, THEN SHRINK BY A FACTOR OF 2

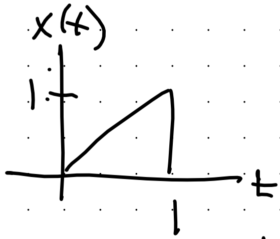


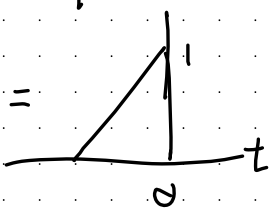
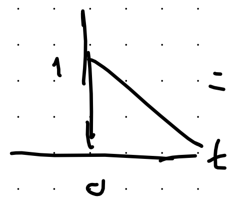
e) $x(2-t) = x(-t+2)$ ← SHIFT LEFT BY 2, THEN FLIP



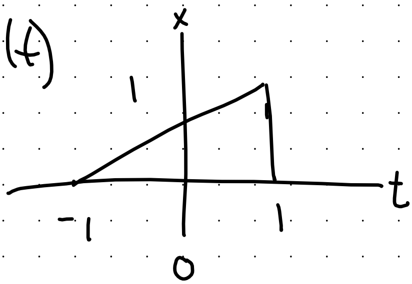
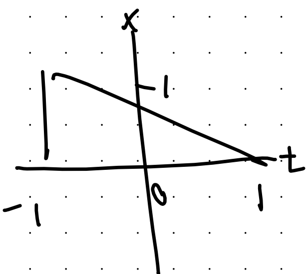
1.2-3

→ WE CAN APPLY OPERATIONS BUT ALSO ADD UP MODIFIED VERSIONS OF $x(t)$



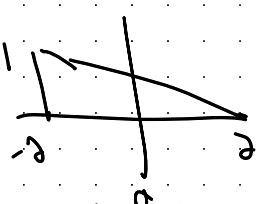
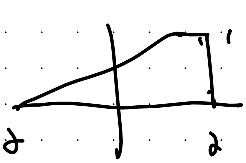
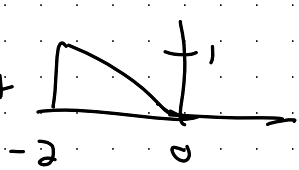
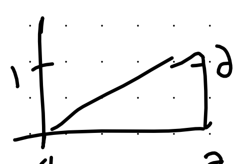
$x_1(t) =$  $+$  $= x(t+1) + x(-t+1)$

$x(t+1)$ $\qquad$ $x(-t+1)$

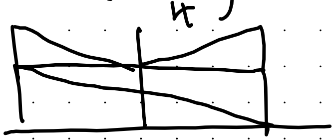
$x_2(t)$  $+$  $= x\left(\frac{t+1}{2}\right) + x\left(\frac{-t+1}{2}\right)$

$x\left(\frac{t+1}{2}\right)$ $\qquad$ $x\left(\frac{-t+1}{2}\right)$

$x_3(t)$ ← WE CAN'T JUST ADD A CONSTANT, SO WE NEED 4 PARTS!

 $+$  $+$  $+$ 

$x\left(\frac{2-t}{4}\right)$ $\qquad$ $x\left(\frac{t+2}{4}\right)$ $\qquad$ $x(-t/2)$ $\qquad$ $x(t/2)$



$$x_3(t) = x\left(\frac{2-t}{4}\right) + x\left(\frac{t+2}{4}\right) + x\left(-t/2\right) + x\left(t/2\right)$$

1-2.3

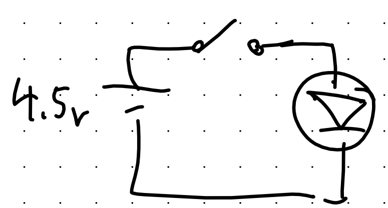
$$x_4(t) = \frac{4}{3}\left[x\left(\frac{t+2}{2}\right) + x\left(\frac{2-t}{2}\right)\right] - \frac{1}{3}\left[x\left(\frac{t+2}{4}\right) + x\left(\frac{2-t}{4}\right)\right]$$

$$x_5(t) = x(t+0.5) + x(0.5-t) + x(t+1.5) + x(1.5-t)$$

1.3-1

THERE ARE OBVIOUSLY AN INFINITE # OF POSSIBLE ANSWERS ...

LET'S CONSIDER A CIRCUIT THAT CONNECTS AN LED TO THREE AA BATTERIES IN SERIES (AND FORGET THE CURRENT LIMITING RES.) WITH A PUSH BUTTON SWITCH. IF WE PRESS IT AT $t=0$ AND RELEASE AT $t=1$



$$v(t) = 4.5\,(u(t) - u(t-1))$$

$v(t)$ IS a) CONTINUOUS, b) ANALOG, c) APERIODIC, d) ENERGY, e) CAUSAL, f) DETERMINISTIC

→ NOT POSSIBLE TO CREATE AN OPPOSITE FOR ALL SIX IN THE REAL WORLD

→ REAL SIGNALS MUST BE FINITE IN LENGTH AND HAVE FINITE ENERGY

→ REAL WORLD SIGNALS ARE FINITE, SO CAN'T BE PERIODIC

→ A SIGNAL COULD BE DISCRETE, DIGITAL, NONCAUSAL, AND RANDOM

→ AVG. YEARLY TEMP, ROUNDED TO THE NEAREST DEG. FROM 1000 BC TO 2000 AD

1.3-3

a) FALSE. FIG 1.11b PG. 80 IS CT BUT DIGITAL

b) FALSE. FIG 1.11c IS DT BUT ANALOG

c) FALSE. e^{-t} IS NEITHER AN ENERGY NOR POWER SIGNAL

d) FALSE. $e^{-t}u(t)$ HAS INFINITE DURATION BUT IS AN ENERGY SIGNAL

e) FALSE. $u(t)$ IS A POWER SIGNAL THAT IS CAUSAL.

f) TRUE. IF IT REPEATS FOR ALL t, THEN IT WOULD HAVE TO HAVE SOME VALUE FOR $t > 0$ AND CAN'T BE ANTI-CAUSAL.

ENERGY SIGNAL: HAS FINITE ENERGY

POWER SIGNAL: HAS FINITE AND NON-ZERO POWER.

REMEMBER

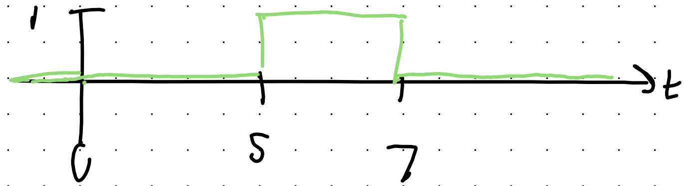
$$P_x = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, dt$$

1.4-1

a) $u(t-5) - u(t-7)$

TURN ON AT $t=5$

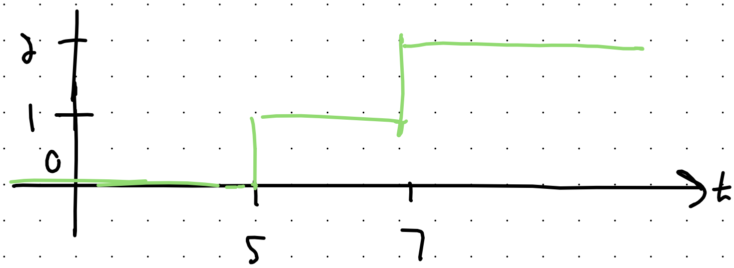
TURN OFF AT $t=7$



b) $u(t-5) + u(t-7)$

TURN ON AT $t=5$

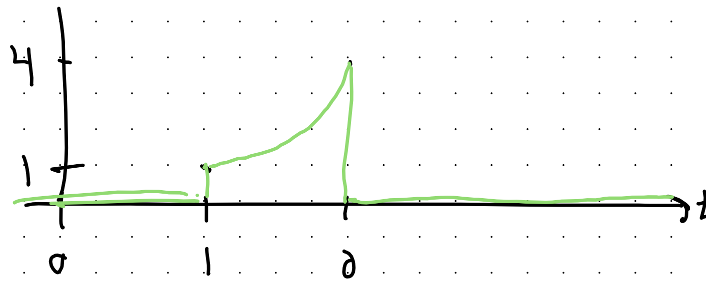
TURN ON AT $t=7$



c) $t^2\left[u(t-1) - u(t-2)\right]$

ON AT $t=1$

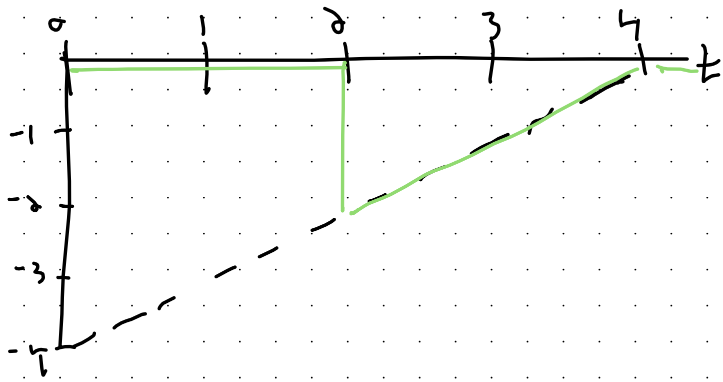
OFF AT $t=2$



d) $(t-4)\left[u(t-2) - u(t-4)\right]$

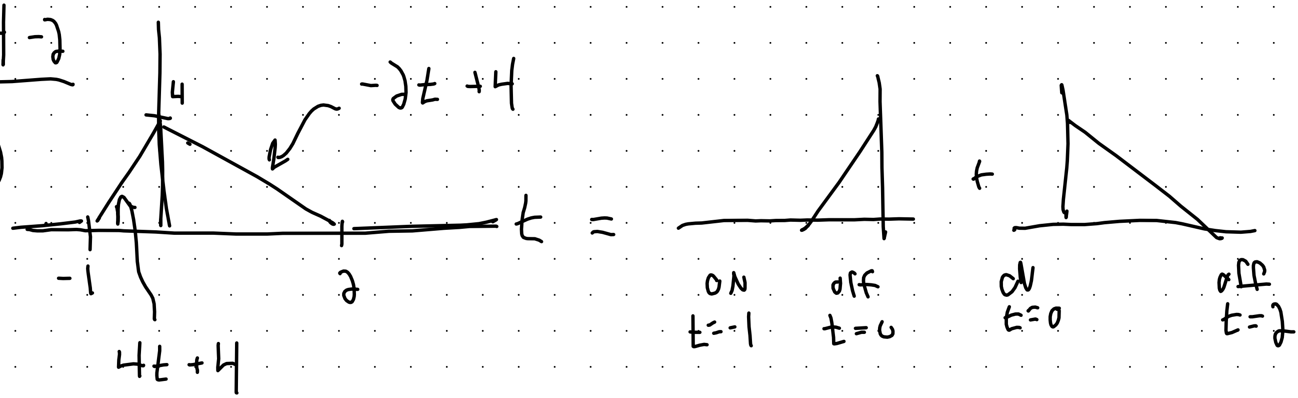
ON AT $t=2$

OFF AT $t=4$



1.4-2

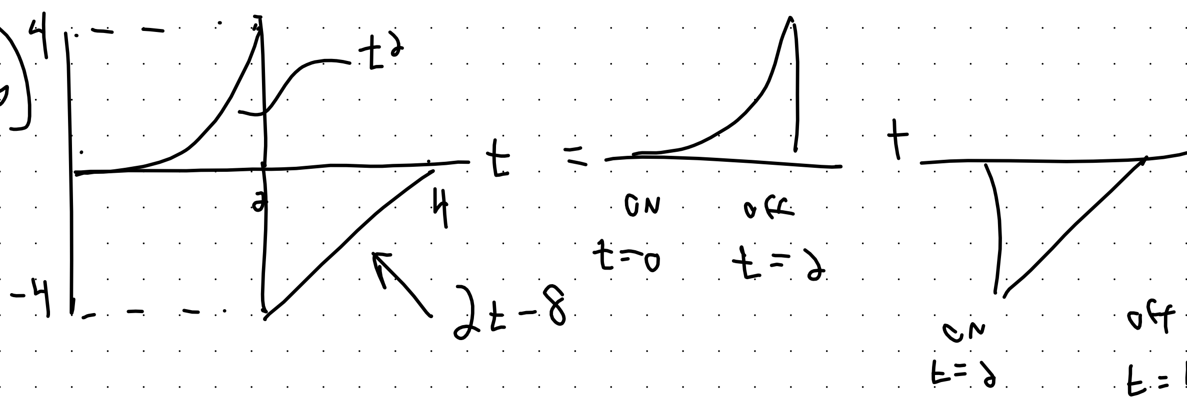
a)



$$x(t) = (4t+4)\left[u(t+1) - u(t)\right] + (-2t+4)\left[u(t) - u(t-2)\right]$$

WHICH CAN BE SIMPLIFIED ... IF YOU WANTED ...

b)



$$x(t) = t^2\left[u(t) - u(t-2)\right] + (2t-8)\left[u(t-2) - u(t-4)\right]$$

WHICH COULD BE EXPANDED AND SIMPLIFIED ...

$$= t^2 u(t) - (t^2 - 2t + 8)\, u(t-2) - (2t-8)\, u(t-4)$$

1.5-1

EVEN AND ODD FUNCTIONS FOR $x(t)$

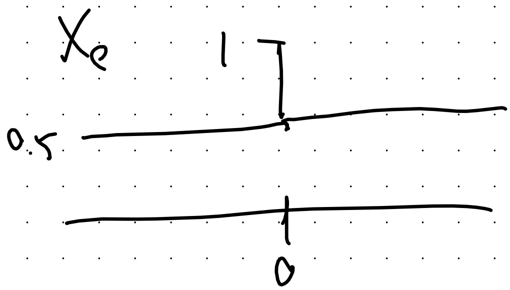
$$x_e(t) = \tfrac{1}{2}\left[x(t) + x(-t)\right]$$

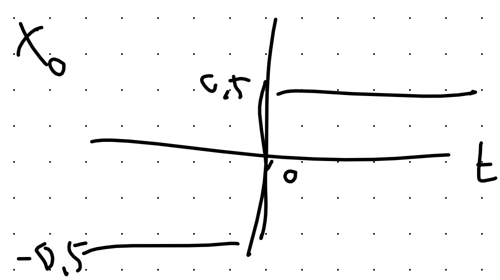
$$x_o(t) = \tfrac{1}{2}\left[x(t) - x(-t)\right]$$

a) $x(t) = u(t)$

$$x_e(t) = \tfrac{1}{2}\left[u(t) + u(-t)\right] = \begin{cases} 0.5 & t \neq 0 \\ 1 & t = 0 \end{cases}$$

$$x_o(t) = \tfrac{1}{2}\left[u(t) - u(-t)\right] = \begin{cases} 0.5 & t > 0 \\ 0 & t = 0 \\ -0.5 & t < 0 \end{cases}$$



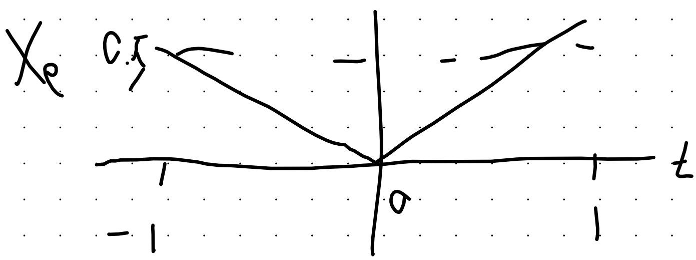


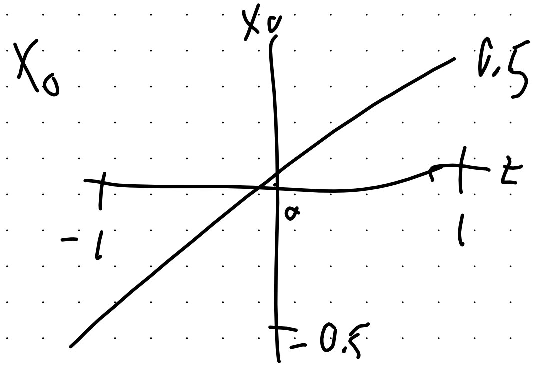
1.5-1

b) $t\,u(t)$

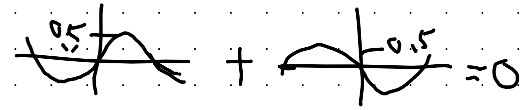
$x_e(t) = 0.5[t\,u(t) + (-t)(u(-t))] = 0.5[t\,u(t) - t\,u(-t)] = 0.5|t|$

$x_o(t) = 0.5[t\,u(t) - (-t)u(-t)] = 0.5[t\,u(t) + t\,u(-t)] = 0.5t$





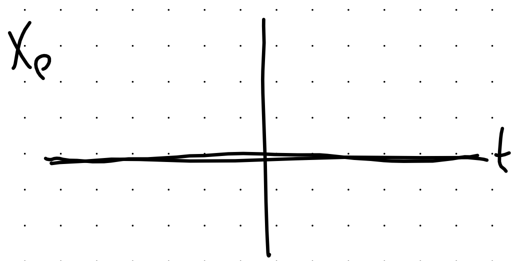
c) $x(t) = \sin(w_0 t)$

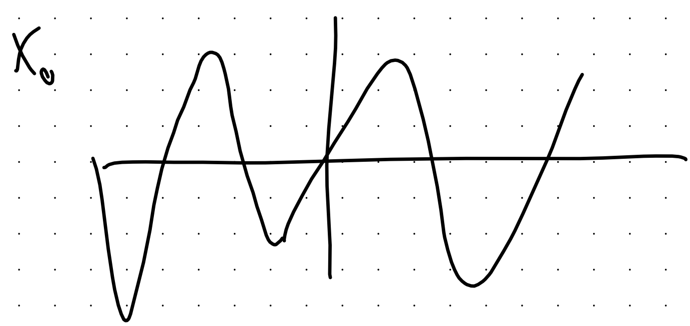


$X_e(t) = 0.5(\sin(w_0 t) + \sin(w_0(-t))) = 0$

$X_o(t) = 0.5(\sin(w_0 t) - \sin(w_0(-t))) = \sin w_0 t$

ONLY ODD PARTS





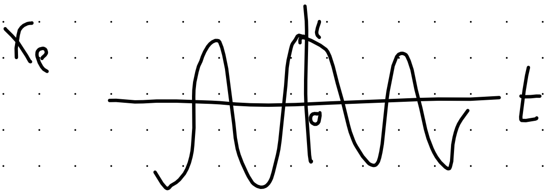
1.5-1

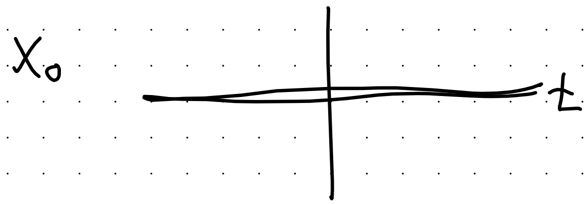
d) $x(t) = \cos(\omega_0 t)$

ONLY EVEN PARTS

$$x_e(t) = 0.5\left[\cos(\omega_0 t) + \cos(\omega_0(-t))\right] = \cos\omega_0 t$$

$$x_o(t) = 0.5\left[\cos(\omega_0 t) - \cos(\omega_0(-t))\right] = 0$$





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e) $x(t) = \cos(\omega_0 t + \theta)$

YOU COULD SOLVE THIS THE LONG WAY, OR USE A TRIG IDENTITY...

$$\cos(\omega_0 t + \theta) = \underbrace{\cos\omega_0 t \cos\theta}_{\text{ONLY EVEN PIECES}} - \underbrace{\sin\omega_0 t \sin\theta}_{\text{ONLY ODD PIECES}}$$

$$x_e(t) = \cos\omega_0 t \cos\theta$$

$$x_o(t) = -\sin\omega_0 t + \sin\theta$$

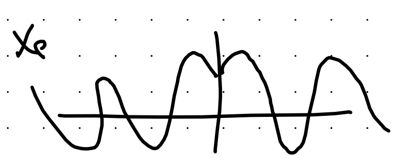
$$x(t) = x_e(t) + x_o(t)$$

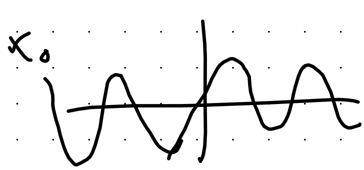
f) $x(t) = \sin(\omega_0 t)u(t)$

FOR THESE YOU HAVE TO RECOGNIZE THINGS WILL CANCEL OUT! TRY DRAWING THEM AS PIECES TO HELP!

$$x_e(t) = \tfrac{1}{2}\left[\sin(\omega_0 t)u(t) + \sin(\omega_0(-t))u(-t)\right]$$

$$x_o(t) = \tfrac{1}{2}\left[\sin(\omega_0 t)u(t) - \sin(\omega_0(-t))u(-t)\right] = 0.5\sin\omega_0 t$$





1.5-1

g) $\cos(w_0 t)\, u(t)$

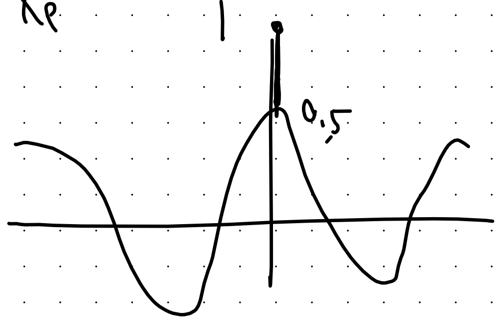
$$x_e(t) = 1/2 \left[\cos(w_0 t) u(t) + \cos(w_0(-t)) u(-t)\right]$$

$$= \begin{cases} 0.5 \cos w_0 t & t \neq 0 \\ 1 & t = 1 \end{cases}$$

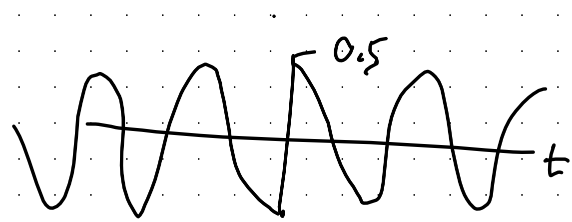
$$x_o(t) = \frac{1}{2}\left[\cos(w_0 t) u(t) - \cos(w_0(-t) u(-t)\right]$$

$$= 0.5 \left[\cos w_0 t\, u(t) - \cos w_0 t\, u(-t)\right]$$

x_e



x_o



1.5-3

a) $x(t) = e^{-2t} u(t)$

$$x_e(t) = \frac{1}{2}\left[e^{-2t}u(t) + e^{-2(-t)}u(-t)\right]$$

$$= \frac{1}{2}\left[e^{-2t}u(t) + e^{2t}u(-t)\right]$$

$$x_o(t) = \frac{1}{2}\left[e^{-2t}u(t) - e^{-2(-t)}u(-t)\right]$$

$$= \frac{1}{2}\left[e^{-2t}u(t) - e^{2t}u(-t)\right]$$

b) $E_x = E_{x_e} + E_{x_o}$

$$E_{x_e} = \int_{-\infty}^{\infty} x_e(t)^2\,dt = \left(\frac{1}{2}\right)^2\left(\int_0^{\infty} e^{-4t}\,dt + \int_{-\infty}^{0} e^{4t}\,dt\right) = \frac{1}{8}$$

$$E_{x_o} = \int_{-\infty}^{\infty} x_o(t)^2\,dt = \left(\frac{1}{2}\right)^2\left(\int_0^{\infty} e^{-4t}\,dt - \int_{-\infty}^{0} e^{4t}\,dt\right) = \frac{1}{8}$$

$$E_{x_e} + E_{x_o} = \frac{1}{4}$$

$$E_x = \int_{-\infty}^{\infty} (e^{-2t}u(t))^2\,dt = \int_0^{\infty} e^{-4t}\,dt = \frac{1}{4}$$

$\therefore E_x = E_{x_e} + E_{x_o}$

1.7-1 CHECK FOR LINEARITY! ADDITIVITY + SCALING!

$x(t)$ INPUT, $y(t)$ OUTPUT

a) $$\frac{dy(t)}{dt} + 2y(t) = x^2(t)$$

CHECK SCALING ONLY

LET $x_1(t) = k\,x(t)$

$$\frac{d}{dt}y_1(t) + 2y_1(t) = x_1^2(t)$$

$$\frac{d}{dt}y_1(t) + 2y_1(t) = k^2x^2(t) = k^2\left(\frac{d}{dt}y(t) + 2y(t)\right)$$

FOR THIS TO BE TRUE: $y_1(t) = k^2 y(t)$

BUT LINEARITY NEEDS: $y_1(t) = k\,y(t)$

$\therefore$ THE SYSTEM IS NOT LINEAR.

---

b) $$\frac{dy(t)}{dt} + 3t\,y(t) = t^2x(t)$$

CHECK ADD. + SCALING

FOR $x_1(t)$ AND $x_2(t)$

$$\frac{dy_1(t)}{dt} + 3t\,y_1(t) = t^2x_1(t) \quad\Big|\quad \frac{dy_2(t)}{dt} + 3t\,y_2(t) = t^2x_2(t)$$

MULTIPLY BY k_1 | MULTIPLY BY k_2

$$k_1\frac{dy_1(t)}{dt} + 3k_1t\,y_1(t) = k_1t^2x_1(t) \quad\Big|\quad k_2\frac{dy_2(t)}{dt} + 3k_2t\,y_2(t) = k_2t^2x_2(t)$$

ADD BOTH TOGETHER

$$\frac{d}{dt}\left(k_1y_1(t) + k_2y_2(t)\right) + 3t\left(k_1y_1(t) + k_2y_2(t)\right) = t^2\left(k_1x_1(t) + k_2x_2(t)\right)$$

THIS IS JUST $y(t) = k_1y_1(t) + k_2y_2(t)$ AND $x(t) = k_1x_1(t) + k_2x_2(t)$

$\therefore$ SUPERPOSITION HOLDS $\therefore$ THE SYSTEM IS LINEAR

1.7-1

c) $3y(t) + 2 = x(t)$

SPECIAL CASE IF

$x_1(t) = 0 \qquad y_1(t) = \frac{x_1(t) - 2}{3} = -\frac{2}{3}$

IF $x_2(t) = 2x_1(t) = 0 \quad y_2(t) = \frac{2x_1(t) - 2}{3} = -\frac{2}{3}$

SO $\quad x_1(t) \rightarrow y_1(t) = -\frac{2}{3}$

$\quad 2x_1(t) \rightarrow y_2(t) = -2/3$

$\quad y_2(t) \neq 2y_1(t)$ SO SUPERPOSITION DOES NOT HOLD

∴ THE SYSTEM IS NOT LINEAR.

---

d) $\frac{d}{dt}y(t) + y^2(t) = x(t)$

LET $x_1(t) = Kx(t)$

FOR LINEARITY WE NEED $y_1(t) = ky(t)$

SUB IN $x_1(t) = Kx(t)$

$$\frac{d}{dt}y_1(t) + y_1^2(t) = Kx(t) = k\left(\frac{dy(t)}{dt} + y^2(t)\right)$$

FOR THIS TO BE LINEAR $y_1(t) = ky(t)$

BUT SUBBING IN $\frac{d}{dt}ky(t) + (ky(t))^2$ CAUSES THIS

TO FAIL.

∴ THE SYSTEM IS NOT LINEAR.

1.7-1

e) $$\left(\frac{dy(t)}{dt}\right)^2 + 2y(t) = x(t)$$

LET $x_1(t) = k\,x(t)$

$$\left(\frac{dy_1(t)}{dt}\right)^2 + 2y_1(t) = x_1(t) = k\,x(t) = k\left(\left(\frac{dy(t)}{dt}\right)^2 + 2y(t)\right)$$

FOR THIS TO BE LINEAR $y_1(t) = k\,y(t)$

BUT SUBBING IN GIVES

$$\left(\frac{d\,(k\,y(t))}{dt}\right)^2 + 2k\,y(t) \neq k\left(\left(\frac{dy(t)}{dt}\right)^2 + 2y(t)\right)$$

$\therefore$ THE SYSTEM IS NOT LINEAR.

1.7-1

(f) $\frac{dy(t)}{dt} + (\sin t)\, y(t) = \frac{dx(t)}{dt} + 2x(t)$

FOR INPUTS $x_1(t)$ AND $x_2(t)$ WE WOULD HAVE OUTPUTS $y_1(t)$ AND $y_2(t)$

(1) $\frac{dy_1(t)}{dt} + (\sin t)\, y_1(t) = \frac{dx_1(t)}{dt} + 2x_1(t)$

(2) $\frac{dy_2(t)}{dt} + (\sin t)\, y_2(t) = \frac{dx_2(t)}{dt} + 2x_2(t)$

MULTIPLY (1) BY k_1 AND (2) BY k_2 AND THEN ADD $k_1(1) + k_2$

$$\frac{d}{dt}\left(k_1 y_1(t) + k_2 y_2(t)\right) + \sin(t)\left(k_1 y_1(t) + k_2 y_2(t)\right)$$

$$= \frac{d}{dt}\left(k_1 x_1(t) + k_2 x_2(t)\right) + 2\left(k_1 x_1(t) + k_2 x_2(t)\right)$$

IF WE CONSIDER $x(t) = k_1 x_1(t) + k_2 x_2(t)$

$y(t) = k_1 y_1(t) + k_2 y_2(t)$

WE SEE SUPERPOSITION HOLDS

$\therefore$ THE SYSTEM IS LINEAR.

1.7-1

THIS IS GOING TO BE A PROBLEM!

g) $$\frac{dy(t)}{dt} + 2y(t) = x(t)\,\frac{dx(t)}{dt}$$

LET $x_1(t) = k\,x(t)$

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1(t)\,\frac{dx_1(t)}{dt} = k\,x(t)\,\frac{d}{dt}\big(k\,x(t)\big)$$

$$\frac{dy_1(t)}{dt} + 2y_1(t) = k^2\,\underbrace{x(t)\,\frac{d}{dt}x(t)}_{\text{SUB IN}}$$

$$\frac{dy_1(t)}{dt} + 2y_1(t) = k^2\left(\frac{dy(t)}{dt} + 2y(t)\right)$$

FOR THIS TO BE TRUE $y_1(t) = k^2 y(t)$

BUT LINEARITY REQUIRES $y_1(t) = k\,y(t)$

$\therefore$ THE SYSTEM IS NOT LINEAR.

1.7-1

h) $$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$

FOR INPUT $x_1(t)$ AND $x_2(t)$ WE HAVE OUTPUTS $y_1(t)$ AND $y_2(t)$

① $$y_1(t) = \int_{-\infty}^{t} x_1(\tau)\, d\tau$$

② $$y_2(t) = \int_{-\infty}^{t} x_2(\tau)\, d\tau$$

MULTIPLY AND ADD $K_1 ① + K_2 ②$

$$\underbrace{k_1 y_1(t) + k_2 y_2(t)}_{y(t)} = \int_{t=-\infty}^{t} (k_1 \underbrace{x_1(\tau) + k_2 x_2(\tau)}_{x(t)})\, d\tau$$

IMAGINE $x(t) = k_1 x_1(t) + k_2 x_2(t)$ WE SEE THIS GIVES THE OUTPUT $y(t) = k_1 y_1(t) + k_2 y_2(t)$

$\therefore$ THE SYSTEM IS LINEAR.

1.7-2 TIME VARIANT → CHECK WHAT A SHIFT IN THE INPUT DOES TO THE OUTPUT

$$x(t+T) \rightarrow y(t+T) \quad \text{FOR INVARIANCE}$$

A SHIFT TO THE INPUTS RESULTS IN THE SAME SHIFT IN THE OUTPUT.

a) $y(t) = x(t-2)$

LET $x_1(t) = x(t-T)$ WE KNOW $y_1(t) = x_1(t-2)$

SO $y_1(t) = x(t-T-2) = y(t-T)$

SINCE

$$x_1(t) = x(t-T) \rightarrow y_1(t) = y(t-T)$$

THE SYSTEM IS TIME INVARIANT

b) $y(t) = x(-t)$ FOR $T>0$;

FOR INPUT $x_1(t) = x(t-T)$ WE GET OUTPUT:

$$y_1(t) = x_1(-t) = x(-t-T) = x(-(t+T)) = y(t+T)$$

$\therefore$ DELAYED INPUT PRODUCES AN ADVANCED OUTPUT

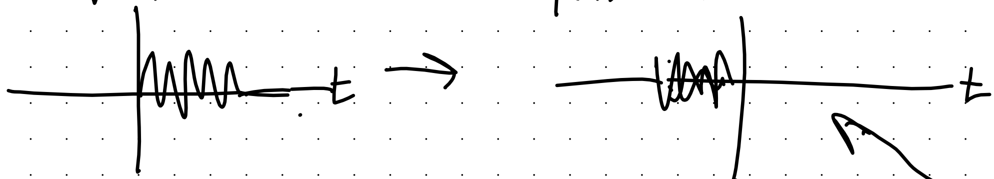
$\therefore$ THE SYSTEM IS TIME VARIANT

1.7-2 b) A GRAPHICAL EXPLANATION

BE CAREFUL, BECAUSE IT SEEMS LIKE IT'S JUST ABOUT SUBBING THINGS IN, BUT IT'S REALLY MORE ABOUT KEEPING TRACK OF WHAT IS HAPPENING.

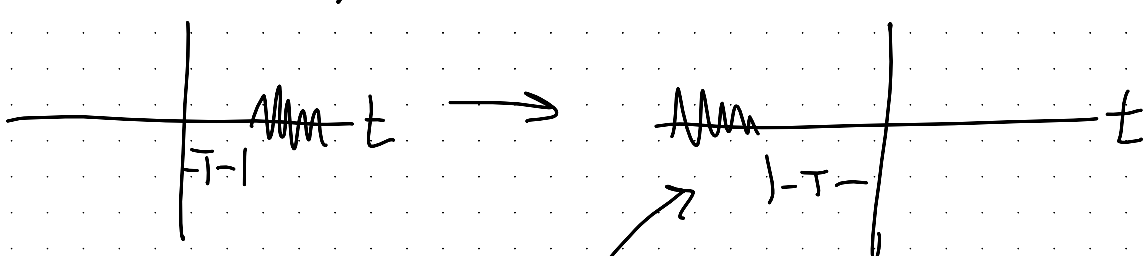
$$x(t) \rightarrow \boxed{A} \rightarrow y(t) = x(-t)$$

$x(t)$ $\qquad\qquad\qquad$ $y(t) = x(-t)$



LETS LET $x_1(t) = x(t-T)$
AND ASSUME $T > 0$

$x_1(t) = x(t-T)$ $\qquad\qquad\qquad$ $y_1(t) = x_1(-t)$



NOW COMPARE $y_1(t)$ IN TERMS OF $y(t)$

$$y_1(t) = y(t+T)$$

BUT FOR TIME INVARIANCE, $y_1(t) = y(t-T)$

$\therefore$ SINCE $y_1(t) \neq y(t-T)$, IT IS TIME VARIANT!

1.7-2

c) $y(t) = x(at)$

FOR $T > 0$, DELAYED INPUT $x_1(t) = x(t-T)$ WE GET:

$$y_1(t) = x_1(at) = x(at-T) = x\left(a\left(t - \frac{T}{a}\right)\right) = y\left(t - \frac{T}{a}\right)$$

A DELAYED INPUT OF T, RESULTS IN A DELAY OF $\frac{T}{a}$

$\therefore$ THE SYSTEM IS TIME VARIANT.

d) $y(t) = t\,x(t-2)$

LET $x_1(t) = x(t-T)$

$$y_1(t) = t\,x_1(t-2) = t\,x(t-T-2)$$

BUT $y(t-T) = (t-T)\,x(t-T-2)$

$$y_1(t) \neq y(t-T)$$

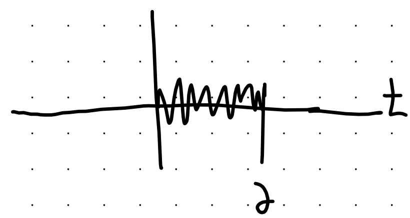
$\therefore$ THE SYSTEM IS TIME VARIANT

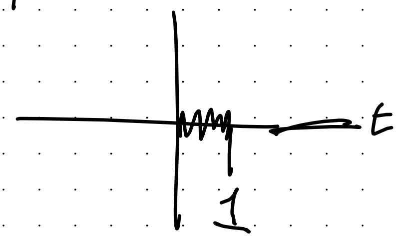
1.7-2 c) GRAPHICAL EXPLANATION!

$$x(t) \rightarrow \boxed{A} \rightarrow y(t) = x(at)$$

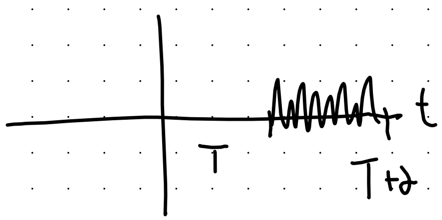
SAY $a = 2$ ← SHRINKING

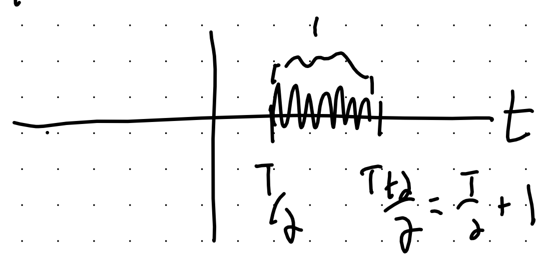
$x(t)$ → $y(t) = x(2t)$





LET $x_1(t) = x(t-T)$ → $y_1(t) = x_1(2t)$





NOW COMPARE $y_1(t)$ TO $y(t)$

$$y_1(t) = y\left(t - \frac{T}{2}\right)$$

BUT FOR INVARIANCE, $y_1(t)$ SHOULD EQUAL $y(t-T)$

$\therefore$ SINCE $y_1(t) \neq y(t-T)$, THE SYSTEM IS TIME VARIANT.

1.7-2

NOTE WE USE τ BECAUSE WE ALREADY USE t IN THE INTEGRATION LIMITS $\int_{t=-5}^{t=5}$

e) $y(t) = \int_{-5}^{5} x(\tau)\,d\tau$ ← $y(t)$ IS A CONSTANT BECAUSE WE ARE INTEGRATING OVER A FINITE TIME PERIOD.

LET $x_1(t) = x(t-T)$

$$y_1(t) = \int_{-5}^{5} x_1(\tau)\,d\tau = \int_{-5}^{5} x(\tau - T)\,d\tau = \int_{-5-T}^{5-T} x(\tau)\,d\tau \neq y(t-T)$$

$y_1(t)$ IS ALSO A CONSTANT, BUT INTEGRATES A DIFFERENT PART OF $x(t)$ WHICH COULD PRODUCE A DIFFERENT VALUE

SINCE $y_1(t) \neq y(t-T)$

∴ THE SYSTEM IS TIME VARIANT

f) $y(t) = \left(\dfrac{dx(t)}{dt}\right)^2$

LET $x_1(t) = x(t-T)$ WE KNOW $y_1(t) = \left(\frac{d}{dt} x_1(t)\right)^2$

$$y_1(t) = \left(\frac{d}{dt}(x(t-T))\right)^2 = y(t-T)$$

SINCE $x_1(t) = x(t-T)$ PRODUCES DELAYED OUTPUT $y_1(t) = y(t-T)$

∴ THE SYSTEM IS TIME INVARIANT

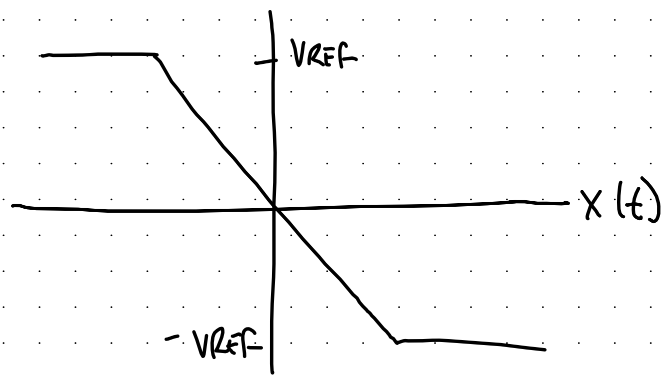
1.7-4

FOR AN INPUT VOLTAGE APPLIED TO AN INVERTING OPAMP

$x(t)$: INPUT

$y(t)$: $$y(t+t_p) = \begin{cases} -V_{REF} & x(t) > V_{REF} \\ V_{REF} & x(t) < -V_{REF} \\ -x(t) & \text{OTHERWISE} \end{cases}$$

WHERE V_{REF} IS THE OP-AMP REFERENCE VOLTAGE (RAILS) AND t_p THE PROPAGATION DELAY ARE BOTH POSITIVE CONSTANTS



a) BOUNDED INPUT BOUNDED OUTPUT

WE SEE THAT $|y(t)| \leq V_{REF} < \infty$ FOR ALL $x(t)$

YES, THE SYSTEM IS BIBO STABLE

1.7-4 CONTINUED

b) CAUSAL?

THE OUTPUT $y(t)$ DEPENDS ON INPUT X AT TIME $t - t_p$ WHICH IS t_p SECONDS IN THE PAST.

NO FUTURE VALUES OF THE INPUT ARE NEEDED TO DETERMINE THE OUTPUT

$\therefore$ THE SYSTEM IS CAUSAL

c) IS IT INVERTIBLE?

PROVE BY COUNTER EXAMPLE...

BECAUSE $x_1(t) = 2V_{REF} \rightarrow y_1(t) = -V_{REF}$

AND $x_2(t) = 3V_{REF} \rightarrow y_2(t) = -V_{REF}$

THE OUTPUT CANNOT UNIQUELY DETERMINE THE INPUT

$\therefore$ NO, THE SYSTEM IS NOT INVERTIBLE

d) IS IT LINEAR $\rightarrow$ CHECK SCALING...

FOR: $x_1(t) = V_{REF} \rightarrow y_1(t) = -V_{REF}$

$x_2(t) = 2V_{REF} \rightarrow y_2(t) = -V_{REF}$

$y_2(t) \neq 2y_1(t)$ FAILS THE SCALING PROPERTY

$\therefore$ NO, THE SYSTEM IS NOT LINEAR

1.7-4 CONT'D

e) MEMORYLESS?

THE CURRENT OUTPUT $y(t)$ REQUIRES A PAST INPUT VALUE $x(t-t_p)$.

NO. THE SYSTEM IS NOT MEMORYLESS.

f) IS THE SYSTEM TIME INVARIANT?

$$x_1(t) = x(t-T) \rightarrow y_1(t) = y(t-T)$$

SYSTEM PARAMETERS DO NOT CHANGE WITH TIME, AND DELAYING THE INPUT CAUSES A CORRESPONDING DELAY IN THE OUTPUT

$\therefore$ YES, THE SYSTEM IS TIME INVARIANT.

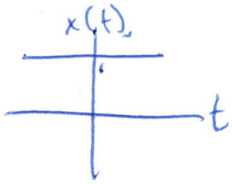
1.7-5

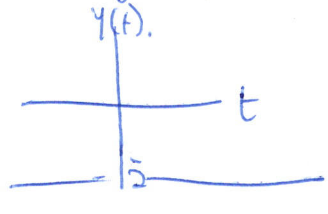
$$y(t+1) = \begin{cases} -2x(t) & \text{WHEN } x(t) \geq 0 \\ 0 & \text{OTHERWISE} \end{cases}$$

IS THE SYSTEM LINEAR?

LET'S CHOOSE A TEST CASE

$$x(t) = 1 \rightarrow \boxed{A} \rightarrow y(t) = -2x(t) = -2$$

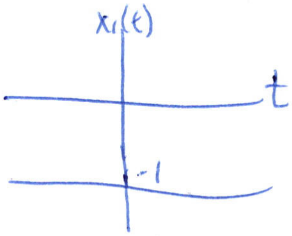




LET'S SCALE BY -1 TO TEST HOMOGENEITY BECAUSE OF THE CONDITION

$$x_1(t) = -1\,x(t)$$
$$x_1(t) = -1$$

$$x_1(t) \rightarrow \boxed{A} \rightarrow y_1(t) = \cancel{-2x_1(t)} = 0$$ BECAUSE $x_1(t) < 0$



$$y_1(t) \neq -1\,y(t)$$

$\therefore$ THE SYSTEM FAILS HOMOGENEITY & IS NOT LINEAR.

---

f) IS THE SYSTEM TIME INVARIANT?

$$x(t) \rightarrow \boxed{A} \rightarrow y(t+1) = \begin{cases} -2x(t) & \text{WHEN } x(t) \geq 0 \\ 0 & \text{OTHERWISE} \end{cases}$$

COMPARE $y(t)$ TO $y_1(t)$

LET $x_1(t) = x(t+T)$

$$x_1(t) \rightarrow \boxed{A} \rightarrow y_1(t+1) = \begin{cases} -2x_1(t) & x(t) \geq 0 \\ 0 & \text{OTHERWISE} \end{cases} = \begin{cases} 2x(t+T) & \text{WHEN } x(t+T) \geq 0 \\ 0 & \text{OTHERWISE} \end{cases}$$

SINCE $y_1(t) = y(t+T)$, THE SYSTEM IS TIME INVARIANT.